

Deep learning based intelligent identification system for ripening stages of banana

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ABSTRACT

Ripening of banana hands during handling, on board transit, shipping and storage leads to higher post-harvest loss and impede the trade. Identification of ripening is paramount importance to reduce loss. Bulk handlers and food processing industries requires automated non-destructive methods of ripening stage identification methodologies. This paper proposes a deep learning based non-destructive method of classification of banana fruit under four categories – unripe, under ripe, ripe and over ripe. A customized dataset was prepared with sufficient images in each class. A convolution neural network (CNN) combined with an eXtreme Gradient Boosting (XgBoost) algorithm (CNN-XgBoost) is introduced for the effective determination of the ripening stage of banana. CNN acts as the trainable feature extractor of the images and XgBoost acts as the identifier of ripening stage. Linear Discriminant Analysis (LDA) is incorporated in order to eliminate the need to have data augmentation or a huge data set. Thus, the proposed deep learning approach possesses capability to perform classification even with a smaller data set compared to conventional deep and machine learning techniques. The performance accuracy of the proposed duo is found to be 91.25 % and it is higher than that obtained with a Support Vector Classifier (SVC), Gaussian Naive Bayesian Classifier (GNB) or k-Nearest Neighbours (KNN) algorithms.

1. Introduction

Ripeness identification of banana is essential in terms of trade, transport, export as well as consumer acceptance. This is also required in the food processing industry to decide upon the type of products to be prepared with the fruit of different ripening stages. India is the largest banana producing country with a production of 31.5 million tons accounting for 20 % of the total world production (NHB, 2022). Banana is one among the most prominent fruit to be exported in the global market in terms of volume and value (20 million tons, 14.5 billion USD) during 2020 (FAO, 2022). Banana also has a fair amount of nutritional value and medicinal benefits (Kumar et al., 2019). Banana offers different health benefits with varying stages of ripening. Green banana fruit (Ripening stage 1–3) has the benefit of controlling blood sugar levels with higher resistant starch (> 45 % dry weight) whereas ripe yellow fruit at the ripening stage 5–6 is good for consumption as fresh fruit because it contains good amount of fruit sugar and provides immediate energy. The over ripe fruit (Ripening stage 7–8) are good for improving

immune system and reduce the risk of certain cancer like breast cancer with high antioxidants (Kyeong et al., 2022; Suresh Kumar et al., 2019). Owing to the high production, India would also likely to be one of the top exporters of banana fruit in the coming years.

Fruit skin color and appearance are standalone factors in terms of consumer acceptance and market price (Mazen and Nashat, 2019). Post harvest loss in banana is ranging from 15 % to 25 % where transport and storage contributes the maximum share (12–20 %). Manual inspection becomes tedious and misleading when handling in large volumes. Banana requires sorting in hands or bunches and not as an individual fruit as normally being practised in oranges, apples, mangoes etc. Hence, world banana fruit market demands fast yet effective non invasive ripening stage dependent classification techniques to enable remote monitoring during shipment, transportation and storage. Sea shipment and other means of export including international supply chain management demands for the identification of the stage of ripening throughout the transit. Efficient non-invasive identification technique during such bulk container handling processes in reefer transport,

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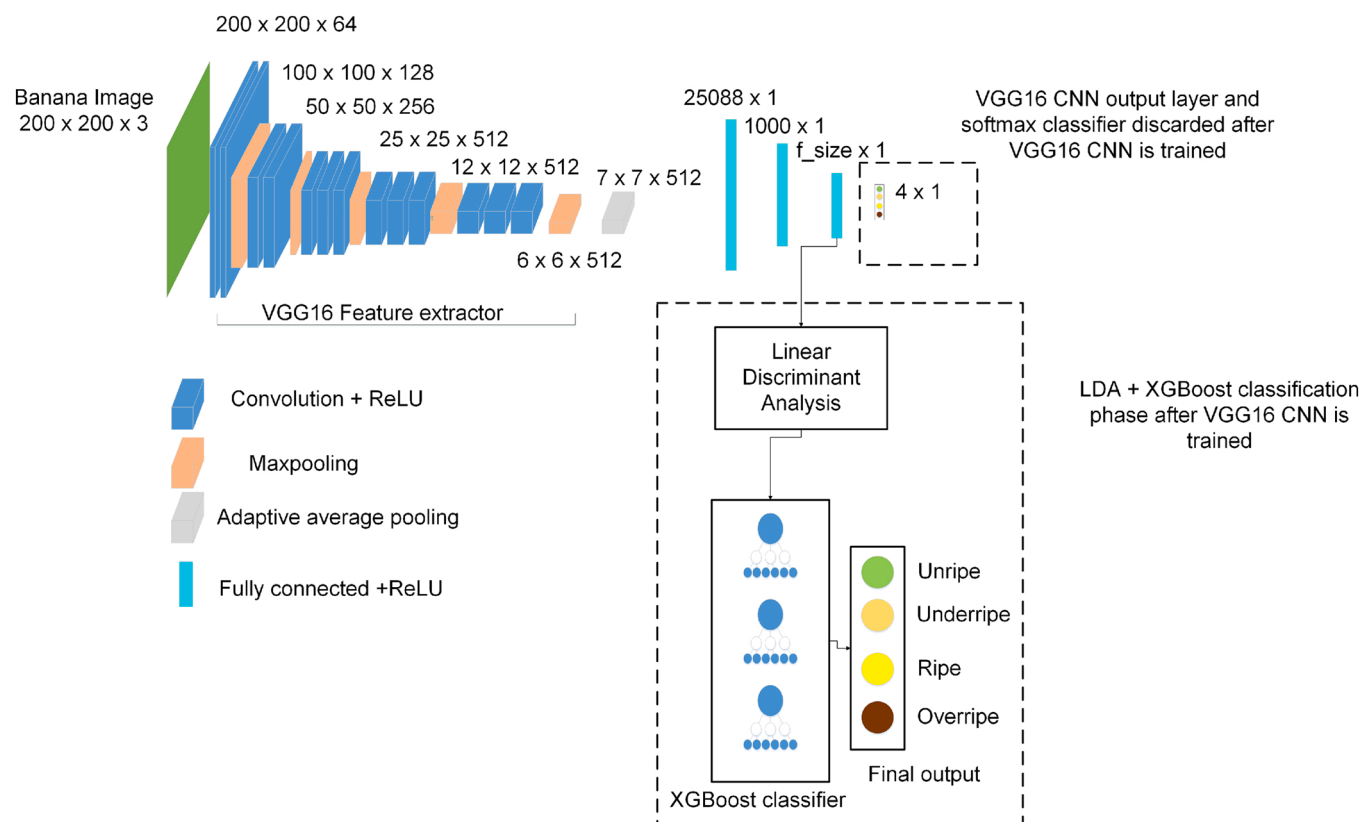


Fig. 1. Architectural block diagram of proposed CNN XgBoost system.

shipping yards, ports, cold storages, ripening chambers and super markets would ease out manual interference.

At the export destinations, the fruit boxes are placed in the ripening chamber and uniformly exposed to ethylene gas (10–100 ppm) according to the ripening cycle. The chamber is maintained at $18 \pm 1^\circ\text{C}$ and relative humidity at 80–85 % for 24 h. Later, the temperature is raised to $20 \pm 2^\circ\text{C}$ and kept for 24 h for the fruits to completely ripen. The ability of automatic identification of the ripening stages will enable the exporters, food inspectors and traders to have timely on-site inspection of the quality of fruit being exported (Ringer and Blanke, 2021). Apart from the export and bulk storage applications, identification of ripeness stage is important in the food processing sector. Processed products from banana include deep fried chips, juice, wine, vinegar, cookies, candies, wafers, beer, puree etc. All these products are prepared from fruit with varying ripening stages. Adapting to newer mobile/computer based non-invasive technological platforms aid in such scenarios to choose upon the required ripeness of banana.

Extensive research studies are being carried out worldwide on the assessment of maturity and ripening stages of banana fruit (Hu et al., 2015). They could be categorized broadly into two categories based on (i) physico-chemical properties based conventional classification (Galal et al., 2022; Kumar et al., 2023) or (ii) advanced computer based machine vision classification (Hu et al., 2015). While the first category would involve to classify the fruit based on a biochemical indicator such as the content of starch, total sugar, total soluble solids (TSS), acidity etc., the second method involves engineering and algorithmic aspects. Obtaining the direct relationship between the process of ripening and the biochemical properties of the fruit is one of the major interests in the first category of classification (Blankenship et al. 2018; Subedi and Walsh, 2011; Soltani et al. 2010). The second category uses machine vision based image processing techniques. The major striking features of incorporating software based techniques are that they are capable to offer completely automatic and non destructive methodologies for the

classification.

Almost all the recent advancements in the computer science, such as internet of things (IoT), machine learning (ML) and deep learning (DL) have been effectively utilised for sophisticated advances in agriculture, horticulture and postharvest technology. Sorting, grading according to the maturity indices, elimination of damaged produces, quality inspection are some of the avenues where computing techniques could be adopted. Process improvement in terms of using AI and ML algorithms has become the go to choice in order to avoid unwanted manual interference. ML based soft computing solutions could be easily be embedded in chip based processors which are relatively smaller in size. This could also possibly provide cleaner, more-efficient, robust supply chain management system with sustainability and profitability.

Research on various state of art technologies on computer vision aided selection and sorting of banana fruits as well as image processing based maturity identification of banana fruits has been performed by researchers (Dimililer and Olaniyi, 2015; Surya Prabha and Sathesh Kumar, 2015; Mustafa et al., 2008; Bora et al., 2015; Hu et al., 2015; Ringer and Blanke, 2021; Galal et al., 2022; Mohapatra et al., 2017; Zhang et al., 2016). One of the most important aspects of implementing computer vision based systems is the choice of appropriate machine learning (ML) algorithms for classification purpose. Although ML algorithms such as Support Vector Machine (SVM), Principal component analyses (PCA), Artificial Neural networks (ANN) are employed for the classification of fruit (Piedad et al., 2018; Le et al., 2019; (Mazen and Nashat, 2019; El-Bendary et al., 2015), CNN has the benefit that it combines feature extraction and classification (L. Zhang et al., 2018; Y. Zhang et al., 2018; Khaing et al., 2018).

In order to avoid the potential problem of overfitting of a CNN, a combination of a CNN and XgBoost network is proposed for effective classification of banana fruit. The advantage of using XgBoost is that it reduces the computation complexity of CNN and also eliminates the need for pooling layer and fully connected layer of the CNN

Table 1
Custom built in CNN model.

Layer	Output shape	Number of parameters	
		Weight	Bias
Input	(Batch size, 3, 200, 200)		
VGG 16 feature extractor	(Batch size, 512, 7, 7)		
Flatten	(Batch size, 25088)		
Linear	(Batch size, 1000)	25,088,000	1000
ReLU	(Batch size, 1000)		
Drop out	(Batch size, 1000)		
Linear	(Batch size, 30)	30,000	30
ReLU	(Batch size, 30)		
Linear	(Batch size, 4)	120	4
Total parameters		25,119,154	

(Thongsuwan et al., 2021; Jiao et al., 2021; Parkavi et al., 2022). A mandatory need for any DL and ML algorithm to perform well is a data set comprising of huge number of images. However, it is not feasible to always ensure having a huge data set. In such scenarios, data augmentation is conventionally used. Data augmentation is a technique which is used to increase the number of images in training dataset through various techniques. It duplicates the image by shifting, flipping, rotating, brightening and zooming in and out the training images (Mureşan and Oltean, 2018). However, this creates additional pre-processing steps for the DL and ML algorithm to work upon.

It is evident that there is a need for a technique which could operate in a smaller data set without data augmentation with promising better accuracy and precision. Most of the classifying techniques in literature require multiple image pre-processing and filtering steps to have accurate classification. Some recognition systems, require additional sensors and equipments to classify. Above all, most of the classification techniques, performed classification based on individual banana fruit termed as “finger”, which in real time seems to be highly impractical.

In order to address the above mentioned shortcomings, the proposed work is aimed to develop an effective deep learning based duo network of CNN and XgBoost architecture to have robust intelligent sorting system for classification of banana fruit based on ripening stages without the need to have huge annotated data or augmentation technique. The proposed technique is also compared with Support Vector Classifier (SVC), Gaussian Naïve Bayesian Classifier (GNB) and k-Nearest Neighbours (KNN) algorithms.

2. Materials and methods

2.1. Proposed architecture

The proposed duo architecture of CNN-XgBoost is implemented in two phases namely the feature extraction and the boosting classifier stage. The architectural block diagram is shown in Fig. 1.

A CNN model is trained to produce features from an input image of fixed dimensions. Once the CNN training is over, the CNN model is ready to be used as a feature extractor. The final softmax classifier layer is discarded. The intermediate activations at the desired layer are the features for the input image. VGG16 is used as the CNN model which is a pretrained architecture. XgBoost is used in the boosting stage to make popularity predictions, given the features extracted by CNN from input images and classify them into one of the four ripening stages. The architecture of the custom built CNN model is shown in Table 1. VGG16 CNN itself has three stages:- feature extraction, adaptive average pooling and classification. In the proposed CNN model the first two phases are unaltered but the classifier part in VGG16 is modified to suit the ripeness identification problem as per the layers mentioned in Table 1. The feature extractor part of VGG16 has a consistent arrangement of convolution layers of 3×3 filter with a stride 1 and always used same padding and maxpool layer of 2×2 filter of stride 2. The training of the hybrid architecture takes place in two phases - (i) CNN training to build

a feature extractor from input images for banana classification (ii) XgBoost training to build an ensemble boosting classifier of features extracted from CNN. The XgBoost training phase takes place after the complete training of CNN. The trained CNN can effectively produce high level features of a given banana image.

The overall flow chart representing the proposed algorithm during the training and testing phase is given in Fig. 2. The image of the target banana hand is acquired and the first step is to resize the image to a decently smaller size. Deep learning architectures train faster on pictures with smaller dimensions. Moreover the proposed architecture requires all the images to be of same size to perform the matrix operations. The images are read from local storage using open cv. The read image object is then converted to tensors operable by Pytorch. The deep learning framework chosen for this work. The next step is image normalisation, where mean and standard deviation based normalization is performed to ensure that the entire data set distribution is centred and the values are limited between 0 and 1. This normalization ensures that the gradient descent learning converges and exploding of gradients due to high value does not happen. The normalized image is then fed to the CNN model, (i.e) the VGG16 pretrained model. The CNN classification is done by a series of fully connected layers and a soft max layer at the end. Cross entropy is used as the loss function to back propagate the errors of CNN predictions in order to learn the weights of CNN classifier stage. The classifier stage weights are discarded after the CNN training is done. The entire training data set is passed again and the intermediate activations from the VGG 16 feature extractor stage are obtained for continuing further training process with XgBoost,. The activations are transformed to a lower dimension in such a way that the class separability between the four classes of image is adequate. The final XgBoost stage is where the trees are grown guided by the objective function of log loss. It is from this stage, the final predicted ripening stage of banana is obtained.

2.2. CNN training phase

The use of convolution and down sampling in CNN have proved to be more advantageous than simple feed forward multilayer perceptrons (MLP), especially for classification tasks with image data. The convolution operation permits us to do weight sharing. This weight sharing happens because of the use of convolutional filters. The output of convolution operation is feature maps whose depth depends on the number of filters used. Each of these filters has a fixed width and height, which acts as a visual receptor that slides over the image. The weights in the filter are learnable during CNN training. The same fixed filter is used across the input image thus reducing the number of parameters in use compared to a MLP. This weight sharing also gives the CNN its invariance property to the affine transformations in the image which enables us to recognize a particular object in the image in any orientation. Pooling is the process of aggregating the feature mappings. It helps in reducing the size of feature maps but preserving important properties. Max pooling picks up the maximum value in the sub regions whereas the activations in sub regions are averaged in average pooling. The pooling layer acts independently on each slice of the input. It reduces the size of the representation space and thus prevents overfitting. Transfer learning has been performed with VGG16 here. Transfer learning enables to employ weights learnt in a similar task and similar data to be reused without any modifications. It is useful in the proposed architecture particularly because of the availability of less data which is not enough to train a deep network without running into problem of memorization. VGG16 has three sections namely (i) feature extractor (ii) adaptive average pooling (iii) classifier. Ideally the entire classifier stage in VGG16 should be disregarded and the nonlinear activations (features) extracted after the adaptive average pooling stage should be fed as input to XgBoost but the number of activations is too high (25,088) for the amount of data in this proposed work. If fed as such it would lead to overfitting. The activations in classifier stage are passed down into

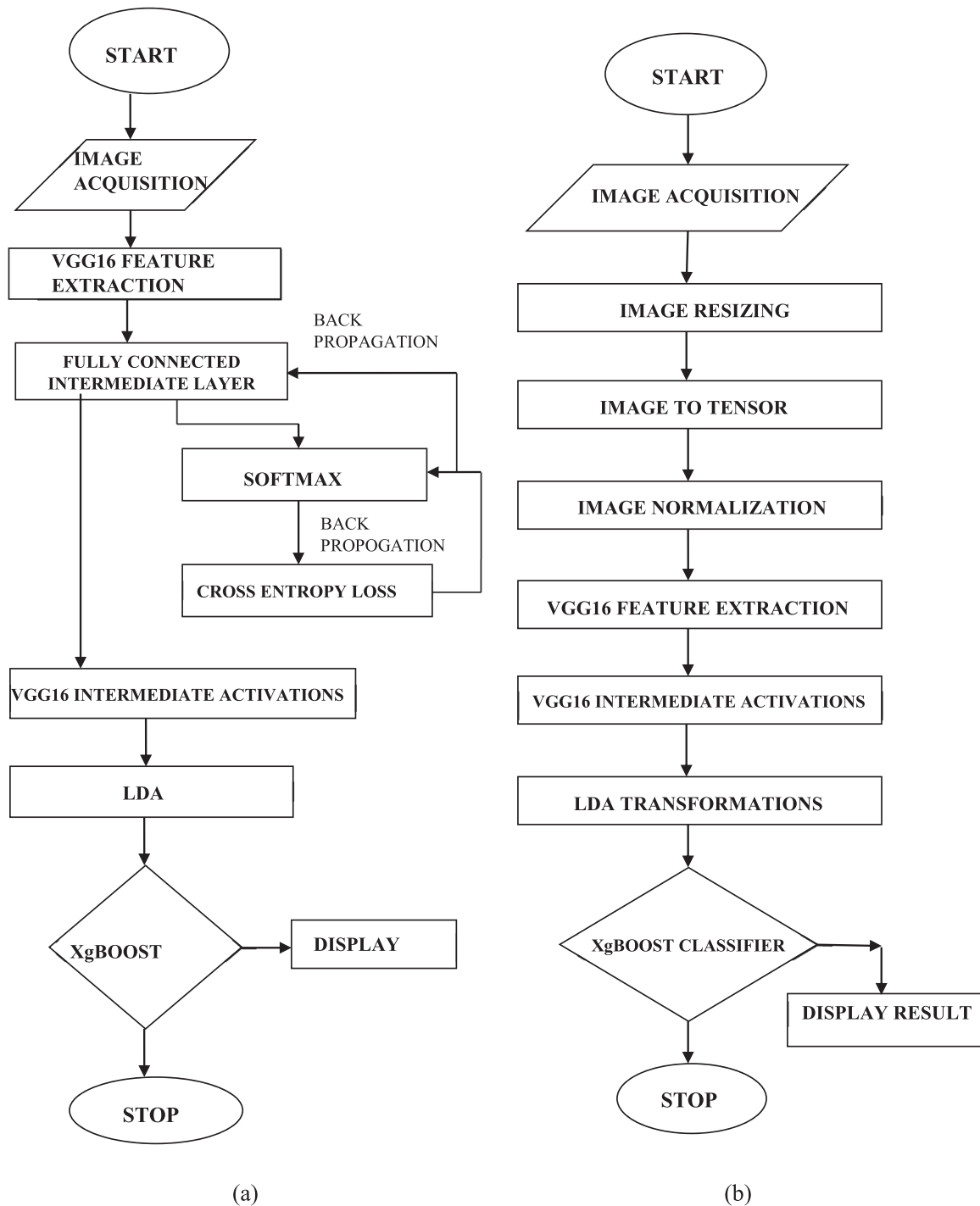


Fig. 2. Flow chart representing (a): Training phase, (b): Testing phase.

further more linear layers with ReLU activation function. The number of activations to be extracted is set as fsz. During training of CNN, the feature extractor weights are frozen and are not modified by back-propagation. Only the weights of linear layers after the feature extractor are trained. XgBoost model is not involved while CNN is being trained. Once CNN training is over, the images are passed once again to the XgBoost layer. The activations at the last layer are extracted and prepared as a data set for XgBoost classifier.

2.3. XgBoost training phase

XgBoost is an optimized gradient boosting algorithm that effectively

uses parallel processing and is highly flexible. XgBoost is basically boosted decision tree which is substituted here for the CNN softmax classifier stage to make final predictions. Decision tree by itself is a popular ML algorithm for classification task but a single decision tree is not sufficient to learn the complex structural interactions of the features particularly in images. Hence a single decision tree is a weak classifier. Boosting is a process in which a weak learner like decision tree is added one by one to fit the residuals of actual probabilities to predicated probabilities. It is fit sequentially in such a way to learn from the mis-classifications made by previous weak learners. Effectively this sequential process builds a strong classifier. After training the CNN, it is used to extract features from the banana images. The entire dataset is



Fig. 3. Sample images from the customised dataset, (R represents ripening stage).

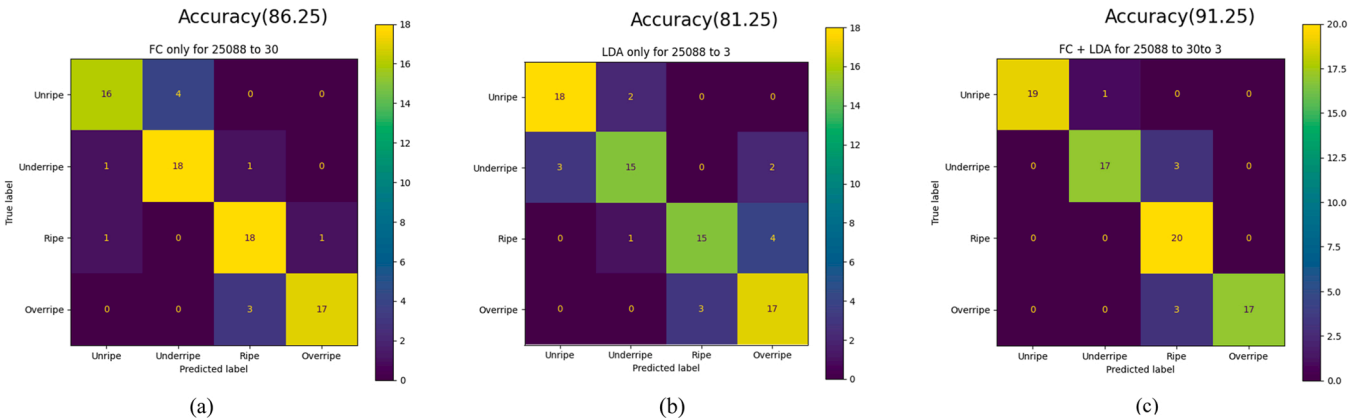


Fig. 4. Confusion matrix of (a) FCN only; (b) LDA only; (c) FCN + LDA.

Table 2
Performance metrics of different cases of dimensionality reduction.

	CNN	CNN FC only + XgBoost	CNN + LDA only + XgBoost	Proposed CNN FC + LDA XgBoost
Accuracy (%)	85	86.25	81.25	91.25
Precision (%)	87.3	87	81.5	92.84
Recall (%)	85	86	81.25	91.25
F1 score (%)	86.13	86	81.4	92.03

again passed through the trained CNN model to extract the activations at penultimate layer of the VGG16 model. The activations along with the labels are collected separately for training, validation and test set. The advantage of using XgBoost as a classifier stage for image prediction is

that it is possible to have an ensemble stage for classifying the extracted features rather than single set of trained classifier weights from CNN classifier stage.

2.4. Linear Discriminant Analysis (LDA)

In spite of regularization techniques added at every stage during CNN and XgBoost training, a degree of over fitting is expected because of high dimensional feature learning with low data. The features learnt from CNN are passed through one more transformation called Linear Discriminant Analysis (LDA) in order to maximize the accuracy of XgBoost. LDA is a supervised dimensionality reduction technique that reduces the high dimensional features to low dimensional features that are more separable in the projected space. The activation is given to XgBoost after applying LDA transformation to make the features more class separable. Basically LDA tries to find a set of axis that would maximize the separation of mean of each class image while minimizing

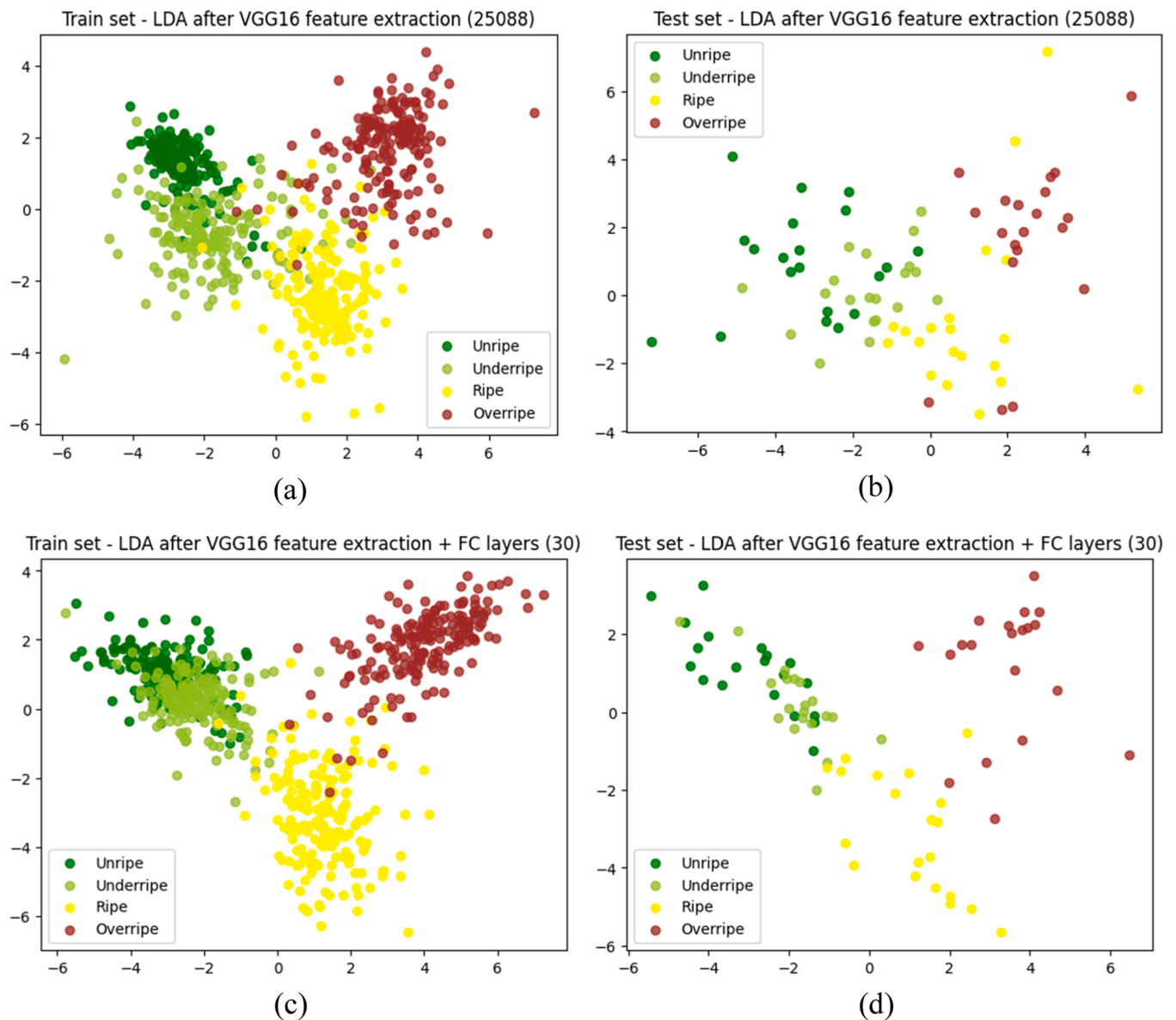


Fig. 5. Scatter plot of LDA (a) on train set after VGG 16 feature extractor; (b) on test set after VGG 16 feature extractor; (c) on train set after VGG 16 feature extractor + Fully connected layers; (d) on test set after VGG 16 feature extractor + fully connected layers.

the variance within each cluster.

2.5. Collection and preparation of customised data set

Banana fruit of different varieties (Grand Naine, Poovan, Ney Poovan, Karpuravalli, Rasthali) were procured from the experimental farm of ICAR - National Research Centre for Banana, Tiruchirappalli, India. Hands, fingers, clusters under four classes namely unripe, under ripe, ripe and over ripe were manually segregated. Complete green fruit (Ripening stages 1–2) were classified as unripe fruit. Greenish yellow colour peel fruit were classified as under ripe (Ripening stages 3–4). Complete yellow peel fruit with few dots were designated as ripe (Ripening stages 5–6). Peel with brown spots, black patches were grouped as over ripe fruit (Ripening stages 7 and above). Photographs of the fruits were taken using Canon Digital SLR Camera (Make: Canon; Model: EOS 200D) under same lighting conditions and the images were saved in.jpg format. In addition, images have also been collected manually from internet. The training data set contained 720 images in total with 180 images in each class with 30 images from the internet.

Validation data set consists of 120 images in total with 30 images in each class out of which 10 images are from internet. Testing dataset consists of 80 images in total with 20 images in each class with 10 images from the internet. Also, these images were selected with care that they represent real practical scenarios. The data set is meticulously collected and prepared to have images that would resemble real world practical scenarios. The dimensions of the images are 6000×4000 . The horizontal resolution is 72 dpi and the vertical resolution is also 72 dpi. The images are collected randomly from each variety without specifically fixing the number of images of each variety. However, it is ensured that almost a balanced number of images were collected representing each variety. Sample images from the data set is represented in Fig. 3.

The idea was to prepare a balanced dataset for the model with all classes having almost equal number of images. An imbalanced dataset may cause the model not to learn the underrepresented class properly and might create reduced accuracy. Although there are sophisticated techniques like adjusting the loss function or providing class weights for class sensitive penalization, a simple approach of generating more data was done. Synthetically dataset for the under ripe class was expanded by

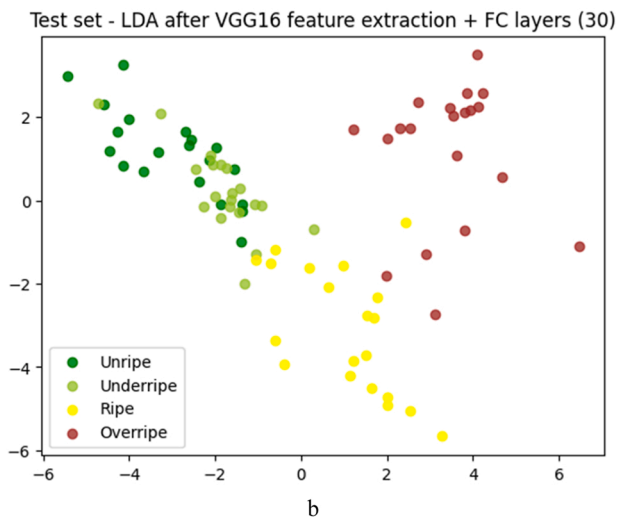
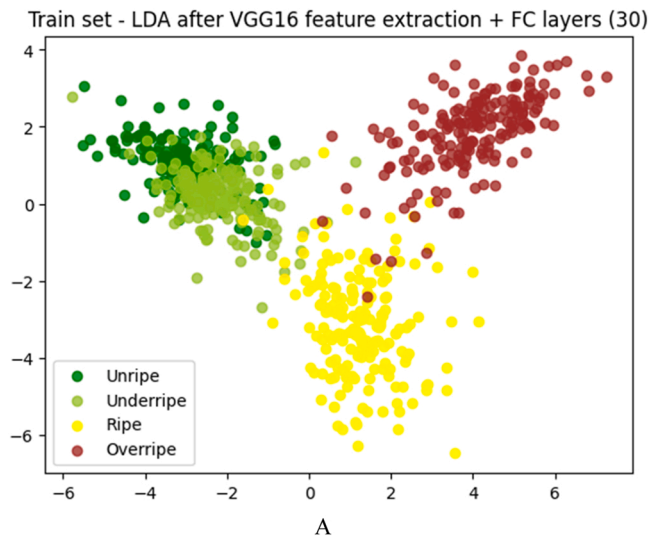


Fig. 6. a Scatter plot of LDA on XgBoost train set. b Scatter plot of LDA on XgBoost test set.

Table 3

Hyperparameter settings of XgBoost-CNN through Bayesian optimization.

Hyperparameters of XgBoost-CNN	Parameter Grid for Bayesian optimization
Learning rate	(0.001, 0.02, "log-uniform")
min_child_weight	(1, 2, 3)
max_depth	(3, 4)
subsample	(0.8, 1, "uniform")
colsample_bytree	(0.5, 1, "uniform")
colsample_bylevel	(0.4, 1, "uniform")
reg_lambda	(1, 5, "log-uniform")
reg_alpha	(1, 10, "log-uniform")
Gamma	(1, 8, "log-uniform")
No. of estimators	(5, 100, "uniform")
Max delta step	(0.1, 0.2)

augmenting the existing images with random transformations which involved sheering, zooming, rotation and brightness changes. Finally, the data set was separated into training, testing and validation folders.

3. Results and discussion

In the present study, CNN is trained with Adam optimizer and batch size is set as 12 and total epochs is set as 25. Cross entropy is used as loss

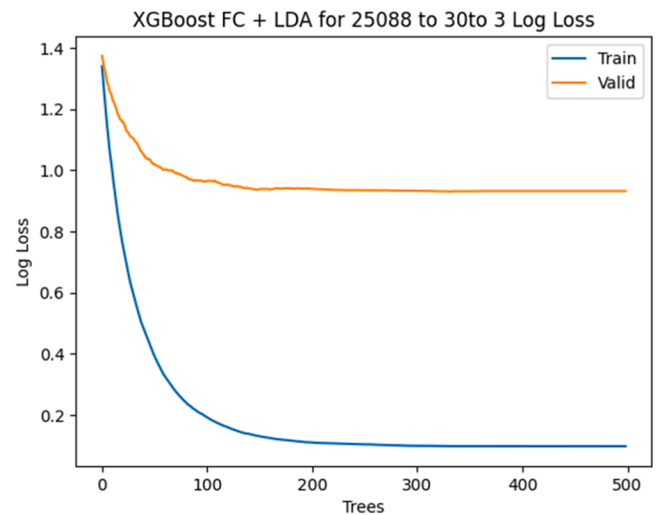


Fig. 7. Loss curve of XgBoost Bayesian LDA.

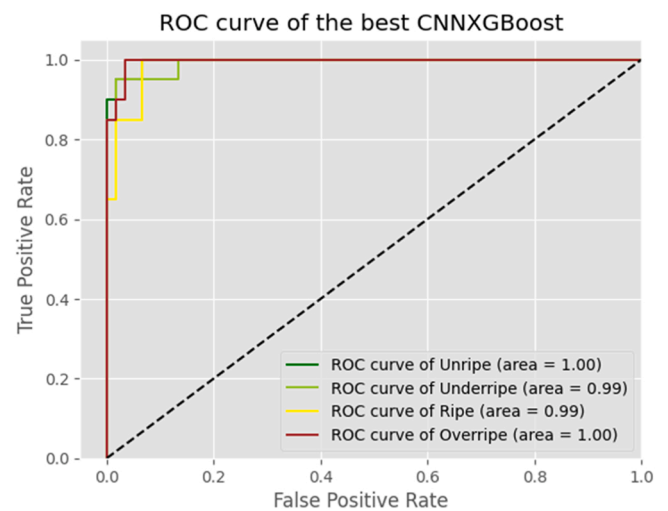


Fig. 8. ROC curve of the proposed CNN XgBoost algorithm for classification of different ripening stages in banana.

function. Learning rate is set as 0.001. Early stopping which monitors the validation set loss is employed to stop training the CNN model when validation loss has not decreased for consequent user defined number of epochs. This is done to prevent high over fitting. Loss curves and accuracy curves are plotted where x axis is number of epochs and y axis is epoch mean loss and epoch mean accuracy over the batches. Other regularization techniques like dropout are employed in the CNN classifier architecture for removing randomly selected neurons from back propagation.

The usage of dimensionality reduction procedures is first analysed. The dimension of features from the VGG 16 is too high for XgBoost to train on directly because of curse of dimensionality. The fully connected layers in CNN could be kept as such or external strategies could be adopted in order to reduce the dimensions either. The classification problem was attempted by using only a fully connected layer, applying LDA directly to 25,088 features and finally the proposed strategy of a combination of both FCN + LDA. This experiment also aligns with the theoretical understanding that CNN requires huge dataset to obtain well learned discriminative features. The inter class similarity is particularly high among under ripe, ripe and unripe bananas which makes it the most challenging class to separate out during testing stage. LDA projects the model learnt activations in a space such that within-class variation is

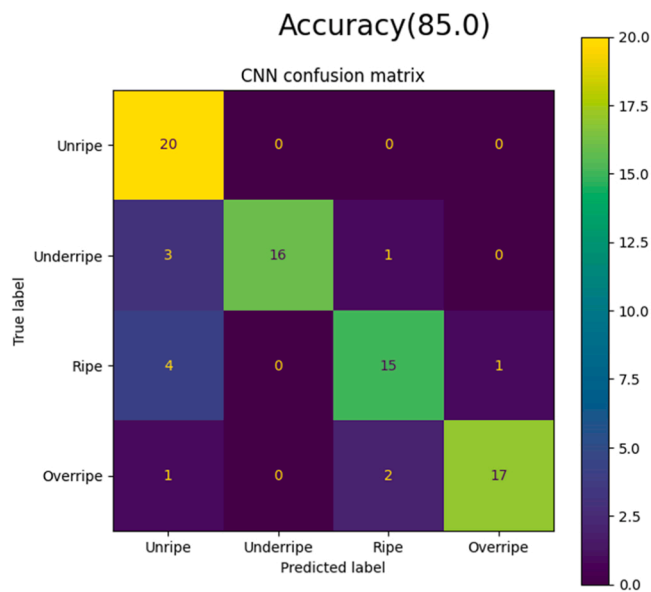


Fig. 9. Confusion matrix of CNN.

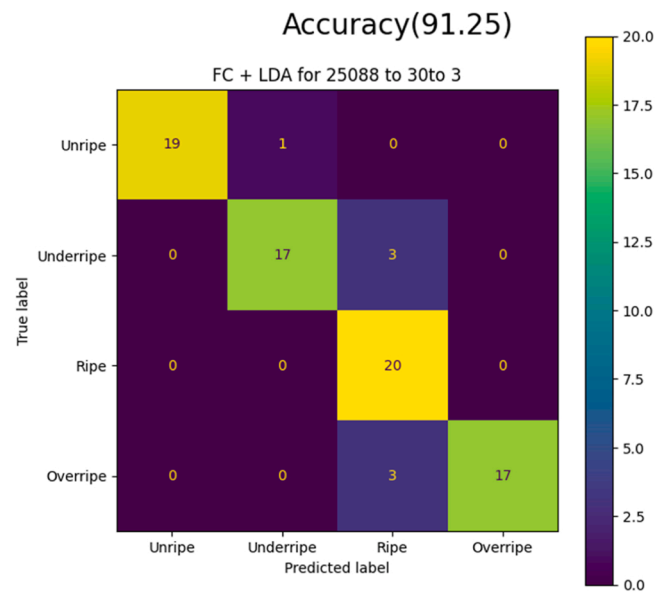


Fig. 11. Confusion matrix of proposed CNN XGB LDA (B).

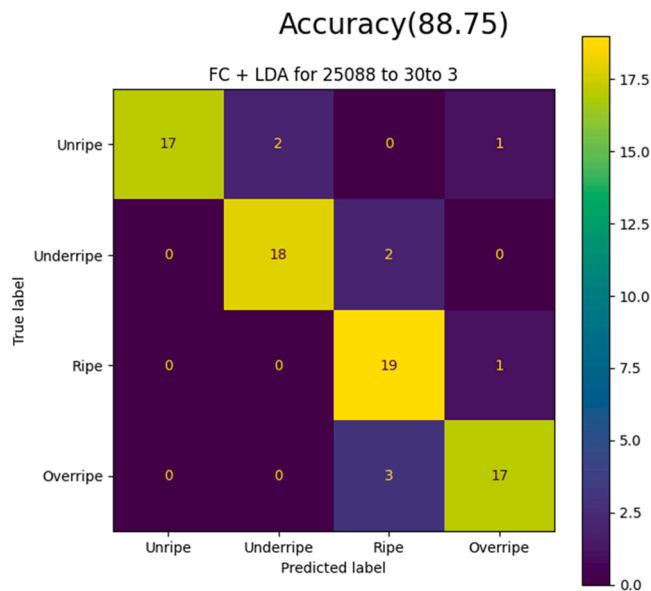


Fig. 10. Confusion matrix of CNN LDA XGB.

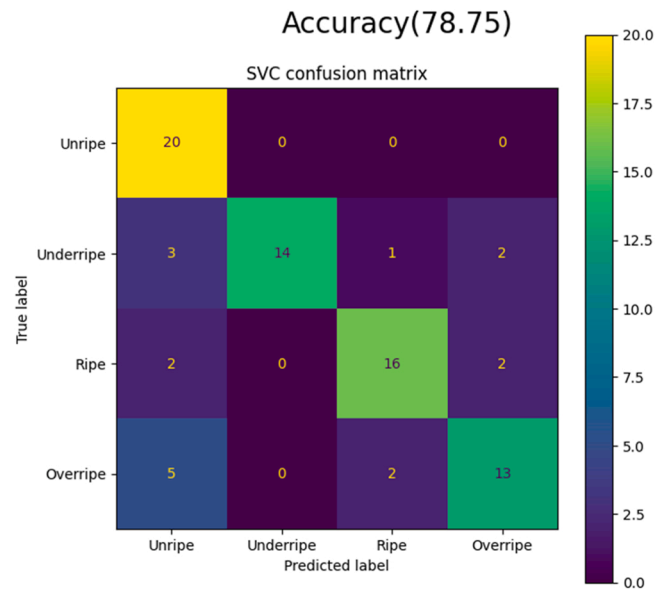


Fig. 12. Confusion matrix of SVC.

small, and the between-class separation is large. When activations were obtained from fully connected layer and given to XgBoost, a decent accuracy score was achieved. With a fixed *fsize* of 30 and input feature dimension of 25088, the following observation on accuracy was made – (i) Accuracy of 82.5 % was obtained with fully connected layer only, (ii) Accuracy of 81.25 % was obtained with LDA fully, (iii) Accuracy of 91.25 % was obtained with fully connected layer & LDA. The confusion matrix of the three cases is presented as given in Fig. 4.

It is evident from Table 2 and confusion matrices Fig. 4 that the combination is a better strategy that provides a good discrimination between the classes.

For visualization purpose the training and testing features obtained from VGG16 which is of size 25,088 are plotted with help of LDA converting it to features of dimension 2. Similarly, it is done for training and testing, after the 25,088 dimensional features undergo feature reduction further in the fully connected layers. It is clear that the inter class separation between features in test set of second case is better than that of

first Fig. 5 case.

The 15-dimensional features from CNN are scaled to standard normal distribution and then transformed to two dimensional feature set using linear discriminant analysis. The features of the training and testing set can be seen in the scatter plots shown in Fig. 6a and b. Each point in the scatter plot is the extracted feature from a single image. It is evident from the scatter plot that highly class wise localised features for the entire dataset has been obtained. It can be seen from the test set that the learned feature space is good enough for XgBoost to identify the different stages of ripening.

The hyper parameters of the XgBoost are tuned via Bayesian optimization algorithm. The tunable hyper parameters of XgBoost that is exposed to the optimization algorithm are number of trees, learning rate, minimum child weight, growth policy, maximum depth of each tree, column sample by tree, by each level of split, ratio of sub-sampled dataset available for each tree, regularization parameters like alpha, lambda and gamma. Skopt. Bayesiansearch cv is used to find the best

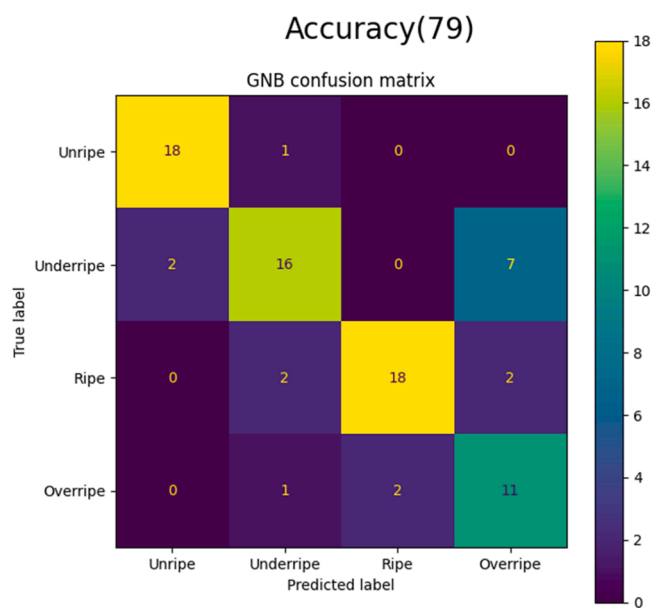


Fig. 13. Confusion matrix of GNB.

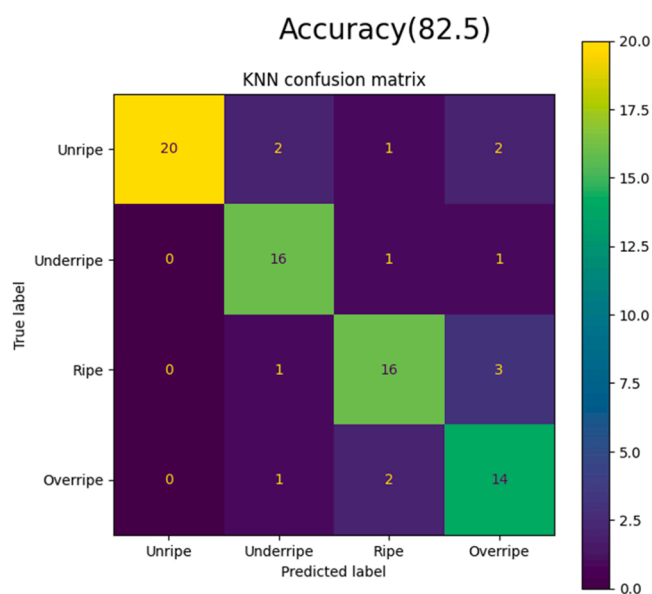


Fig. 14. Confusion matrix of KNN.

Table 4

Performance metrics.

	CNN	CNN LDA XGB (B)	GNB	SVC	KNN
Accuracy (%)	85	91.25	79	78.75	82.5
Precision	87.3	92.84	80	82	83
Recall	85	91.25	79	79	82
F1 score (%)	86.13	92.03	78	79	82

CNN – Convolution Neural Network; CNN LDA XGB (B) – Convolution Neural Network – XgBoost with Bayesian optimization; SVC – Support Vector Classifier; GNB – Gaussian Naive Bayesian Classifier; KNN – k-Nearest Neighbours.

hyper parameters from a grid of values for each parameter. The minimum to an objective function is identified by the Bayesian optimization algorithm by using Gaussian process regression and, hence, it is found to be more suitable for continuous values.

The hyper parameter optimization takes place by setting up a five-fold cross validation scheme for the training set. Basically, the training data is split into 5 parts with class distributions same across all the splits. In each run, one part is set as test set and the remaining parts are set as training sets. Accuracy is passed as the evaluation metric. At each run of the cross validation scheme, Bayesianian optimization chooses parameters that tries to maximize accuracy. The XgBoost hyperparameters are tabulated in Table 3. XgBoost has a number of parameters that needs to be set for this classification task namely the (i) loss function - 'multi: softprob', which is a multi classification loss function that optimizes the softmax probabilities of each class. The output class is the one with the highest probability. (ii) eval set - Validation set to perform early stopping so as to prevent adding more trees if there is no considerable decrease in validation loss. Two scenarios have been validated in this work – (1) default Hyperparameter set for CNN- XgBoost with LDA (2) Hyperparameter through Bayesianian optimization set for CNN- XgBoost with LDA. In both cases Bayesianian optimization has yielded higher accuracy than default XgBoost parameters.

The evaluation metric for the Bayesianian optimization is set as Log loss and from the curve in Fig. 7, it can be observed that there was a decreasing trend in training and validation set.

ROC curve in Fig. 8 is plotted for the best version of the XgBoost classifier whose parameters have been optimized with Bayesian optimization. The Receiver operating characteristic curve shows the performance of a classifier at all classification thresholds. From the ROC graph of CNN XgBoost with LDA, the following points can be inferred.

- CNN has trouble in distinguishing some ripe images from unripe. Also, there exists rapid confusion in identifying under ripe and ripe images.
- CNN XgBoost has overcome this problem and it is good at not misinterpreting other stage images as unripe.

Other than the deep learning models, a comparison was made between the various ML algorithms after doing basic pre-processing steps on the image to extract features. Since the basic distinguishing property among the various ripening stages is related to colour and the texture of the objects in the images, colour and texture features are used. Lab Histogram and attributes of Gray level co-occurrence matrix are concatenated as one feature vector and fed to the ML algorithms. Texture features give information about spatial arrangement of color or intensities. Random patches have been taken from Gray scale image of the banana image and for each patch GLCM (Gray level co-occurrence matrix) properties like dissimilarity, correlation, homogeneity, energy and contrast are used thus creating a 5 dimensional texture feature. i_j^{th} cell in GLCM tells how many times i and j colour value occur together with a certain distance and certain angle between them. Among the deep learning models a comparison has been established between CNN, CNN XgBoost with default and CNN XgBoost parameters optimized with Bayesian optimization.

The confusion matrices of the proposed architecture and the other state of art methods in literature is shown in Figs. 9–14. It is seen that the accuracy of the proposed architecture with Bayesian optimization as shown in Fig. 11 has the highest accuracy of 91.25 %. Also, the false positive value is much reduced than the other algorithms in comparison. The proposed algorithm is still able to classify appropriately even though an unbalanced data set has been used. The ML algorithms have performed poorer than the deep learning models even with a manually crafted pre-processing stage. It is to be noted that all the work spent on finding the best features for the image is automatically handled by the deep learning models where CNN performs the best feature extraction. SVC is a good alternative for neural networks in case of low data. However, it performed poor in this study probably due to a considerable overlap among the images. SVC reports a high precision for overripe images and high recall for unripe images. It is evident that SVC is only capable of separating the dark brown and black tints of the over ripe

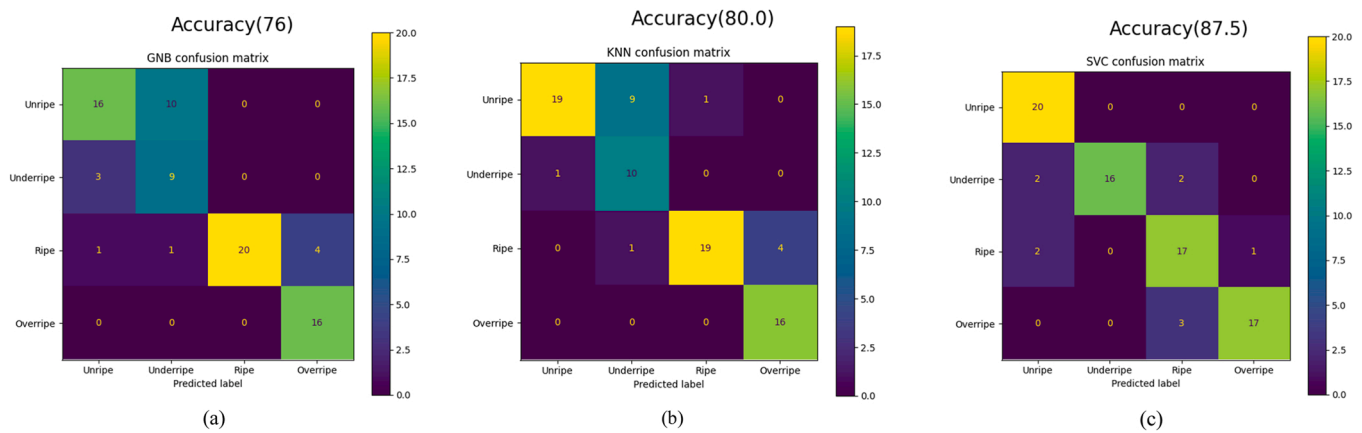


Fig. 15. Confusion matrix of (a) CNN – GNB; (b) CNN – KNN; (c) CNN – SVC.

image from the light green and yellow tints of the other stage.

In order to compare the performance of the proposed method with other state of art methods available in literature, class wise precision and recall were calculated and plotted for all the algorithms under comparison. Precision tells how many items are actually belonging to a particular class among the items reported as belonging to that class by the algorithm. Precision, Recall, Accuracy and F1 score are calculated by the formula specified in Eqs. (1)–(4).

$$\text{Precision} = \frac{\text{True Positive}}{\text{Total predicted positive}} \quad (1)$$

$$\text{Recall} = \frac{\text{True Positive}}{\text{Total actual positive}} \quad (2)$$

$$\text{Accuracy} = \frac{\text{Correct predictions}}{\text{Total predictions}} \quad (3)$$

$$\text{F1score} = 2 \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

On comparing CNN XGB with LDA and Bayesopt, it reports slightly more false positives in case of Ripe and Underripe images which is evident from lower precision compared to CNN but this has been balanced by a higher recall score compared to plain CNN. In addition to the performance evaluation via precision, recall, accuracy and F1 score of the proposed scheme of CNN LDA XgBoost hyperparameters tuned using Bayesian optimization, performance metric of CNN LDA XgBoost with default hyperparameter, and conventional CNN are also presented. Thus, it is evident that the developed model has tried to cover the maximum range of ground truths of ripe and underripe bananas. In a transportation environment a higher recall might be more desirable where maximum correct classifications for a particular stage is required as there would always be secondary stage of checking and the few false positives could be filtered out. It is observed from the accuracy and F1 score in Table 4 that CNN performed better than the ML algorithms. It is confident when it classifies an image as overripe evident from the high precision. This means CNN has learnt features to identify the extremely different class of images even with limited data. It is clearly evident from the results presented, that the proposed CNN-XgBoost algorithm with Bayesian optimization is capable to identify the trained for classes of ripening with a comparatively high accuracy of 91.25 %.

The CNN classifier stage is a series of fully connected layers with progressively decreasing neurons in layer – a funnel shape. This format is chosen to streamline the large dimension features from VGG16 into smaller dimension features for low level processing at the XgBoost stage. A good accuracy of 91.25 with CNN LDA XgBoost in a higher batch size of 128 for CNN is achieved. While for the same batch size, the CNN alone yielded an accuracy of 85. However, a better accuracy of 88.75% was

reached in CNN with batch size set to a low value of 64. Having a higher batch size is, desirable for GPU parallelism purposes and scalability. The stand alone CNN is indeed better but it means that it needs more data and considerably low batch size to get an incremental increase in accuracy. However, with the proposed implementation of replacing the last softmax layer with LDA XgBoost combination, a higher accuracy of 91.25 % is achieved. SVC, GNB and KNN were chosen as a replacement for the XgBoost classifier and the performance was observed for a better comparison purpose.

The features obtained from the combination of CNN fully connected layers and LDA is given to the other three classifiers for comparison with XgBoost as shown in Fig. 15. Out of them only SVC was able to reach a good amount of accuracy. Yet, it is lower than what is observed with XgBoost – 91.25 %. SVC is indeed comparable to XgBoost as it is also performing good in small dataset situation. However, SVC is known to take considerable computation time when dataset size is increased and choosing an appropriate kernel would be a daunting task compared to running a bayes optimization search on the grid mentioned above for XgBoost. Hence considering scalability in mind, XgBoost was chosen as the best option.

The proposed technique is able to achieve a higher precision and accuracy than the other ML techniques available in literatures. Similar to the human vision system, the proposed algorithmic process is able to distinguish between the exact local and global variations in different ripening stages of banana. While other machine learning and deep learning algorithms in literature operates with additional data augmentation, image pre-processing techniques for image classification the proposed technique as discussed eliminates the necessity of major pre-processing steps. The feature extraction is taken care by the inbuilt VGG 16 architecture. Many computer vision based image classification approaches require each pixel information for consideration and then manual feature extraction making the process cumbersome and time consuming. However, by implementation of the proposed CNN-XgBoost duo architecture in the process line, the feature extraction and classification of ripening stages are done automatically in a single step with the pre-trained model. It is to be noted that the proposed architecture is able to operate on a small data set and still produce classification results with higher accuracy.

4. Conclusion

A completely non-invasive technique to identify four classes of banana fruit has been proposed using the duo of CNN-XgBoost with LDA. The experimental validation results of the proposed technique proved its supremacy over other existing machine learning and deep learning algorithms. While the major requirement of deep learning algorithms is to have a sufficiently large data set, the proposed scheme is capable to

provide promising results with comparatively smaller data set. This serves as an initial step for classifying complex and clustered bananas using deep learning in industrial export and processing applications. In addition, the computational complexity of the proposed architecture is comparatively lesser with desired accuracy. This in turn reduces the burden of memory in the processor and hence aiding in effective energy utilisation. This proposed method of non-invasive technique could be used by the warehouse managers, exporters and to monitor the health of bananas on a real time basis. This would also pave way to create mobile application based platforms.

CRedit authorship contribution statement

Shuprajhaa Thayumanavan: Conceptualization, Methodology, Data curation, Writing – original draft. **Mathav Raj:** Software, Data curation, Writing – original draft. **Paramasivam Suresh Kumar:** Data curation, Writing – review & editing, Resources, Funding acquisition, Supervision. **Sheeba K N:** Writing – review & editing. **Uma Subbaraya:** Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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